

Hyperleukocytosis in a neuroblastoma patient after treatment with natural killer T cells expressing a GD2-specific chimeric antigen receptor and IL-15

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To cite: Tian G, Courtney AN, Yu H, *et al.* Hyperleukocytosis in a neuroblastoma patient after treatment with natural killer T cells expressing a GD2-specific chimeric antigen receptor and IL-15. *Journal for ImmunoTherapy of Cancer* 2025;**13**:e010156. doi:10.1136/jitc-2024-010156

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/jitc-2024-010156>).

Accepted 23 November 2024

ABSTRACT

The ability of immune cells to expand numerically after infusion distinguishes adoptive immunotherapies from traditional drugs, providing unique therapeutic advantages as well as the potential for unmanageable toxicities. Here, we describe a case of lethal hyperleukocytosis in a patient with neuroblastoma treated on phase 1 clinical trial (NCT03294954) with autologous natural killer T cells (NKTs) expressing a GD2-specific chimeric antigen receptor and cytokine interleukin 15 (GD2-CAR.15). This patient was the first to be treated on dose level (DL) 5 and the first patient whose product was restimulated with K562-derived artificial antigen-presenting cells (aAPCs) instead of autologous peripheral blood mononuclear cells (PBMCs). 12 previously treated patients on DLs 1 through 4 did not experience significant toxicity. Our root-cause analysis revealed no genetic alterations of known clinical significance and excluded the possibility of clonal expansion due to insertional retroviral mutagenesis. We report that the use of aAPCs instead of PBMCs for CAR-NKT restimulation contributed to a hyperproliferative state associated with distinct gene expression that possibly led to explosive lymphocyte expansion and uncontrolled toxicity in the patient. These findings warrant the implementation of measures to control immune cell activation during manufacture of cell therapy products, especially those armed with transgenic cytokines.

INSIGHTS

We report a case of lethal non-clonal hyperleukocytosis in a neuroblastoma patient treated on a phase 1 trial (NCT03294954) with autologous natural killer T cells (NKTs) expressing a GD2-specific chimeric antigen receptor and IL-15 (GD2-CAR.15) that may have been caused in part by overstimulation of NKTs with K562-based artificial antigen-presenting cells during manufacture.

INTRODUCTION

Neuroblastoma (NB) is one of the most common and deadly pediatric solid tumors

with half of patients presenting as high-risk and only half of these cases remaining in remission long-term. The GD2 disialoganglioside is expressed at high levels in most NB tumors and has been validated as a therapeutic target for antibody-mediated immunotherapy.¹ GD2-specific CAR T cells have also been shown to mediate objective responses without causing significant toxicity in early-stage clinical trials.² In a recent interim analysis, we described the first clinical evaluation of natural killer T cells (NKTs), a sub-lineage of innate-like T cells, engineered to express a GD2-CAR and cytokine interleukin (IL)-15 (GD2-CAR.15) to treat children with relapsed/refractory NB (GINAKIT2, NCT03294954, online supplemental file 5).³ Results from the first 12 patients in this trial showed that GD2-CAR.15 NKTs are well tolerated and mediate a 25% objective response rate in heavily pretreated NB patients receiving up to 1×10^8 GD2-CAR.15 NKTs/m² at dose level (DL) 4.³ Subsequently, the first patient treated on DL5 (3×10^8 GD2-CAR.15 NKTs/m²) developed hyperleukocytosis with resulting multi-organ failure and eventual death. Here, we describe clinical and pathological manifestations of this unusual toxicity and present a root-cause analysis at the cellular and molecular level.

CASE DESCRIPTION

Patient P5-1 (DL5-patient 1) had relapsed/refractory, high-risk stage M (4) NB and underwent induction and consolidation for upfront therapy followed by four salvage regimens for refractory NB prior to enrollment on the GINAKIT2 study. At the time



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of enrollment, this patient's disease involved multiple cranial bone lesions and retroperitoneal mass with a Curie score of 3. After meeting eligibility criteria, the patient was enrolled on DL5 and underwent lymphodepletion with cyclophosphamide and fludarabine followed by infusion of GD2-CAR.15 NKTs on day 0.

Patient P5-1 was admitted from day 0 to day 4 for fever that improved with supportive management ([figure 1A](#)); this admission was attributed to metapneumovirus infection unlikely related to GD2-CAR.15 NKTs. The patient was re-admitted on day 5 for fever that was initially managed with supportive measures. On day 7, tocilizumab was administered then anakinra on day 8 for cytokine release syndrome (CRS). The patient became afebrile on day 12 but exhibited nystagmus-like eye movements with EEG negative for seizure on day 11. Brain MRI showed enlargement of bony NB metastases concerning for progression versus pseudo-progression. On day 13, patient P5-1 developed strabismus and involuntary upward gaze and was started on 0.2mg/kg q6h dexamethasone intravenously for possible immune effector cell-associated neurotoxicity syndrome.

On day 14, a significant increase in white cell count was noted ([figure 1B](#)) with drastic increases in absolute lymphocyte counts and ferritin level ([figure 1C,D](#)). C reactive protein level was initially elevated and decreased after tocilizumab and anakinra treatment ([figure 1E](#)). By day 16, the patient developed symptoms resembling sinusoid obstruction syndrome of the liver (attributed to leukostasis) with fluid retention, weight gain, and decreased renal function. On day 17, patient P5-1 was transferred to the intensive care unit due to progressively worsening hyperleukocytosis. The patient's overall condition rapidly deteriorated, requiring intubation for mechanical ventilation and blood pressure support with epinephrine, norepinephrine, and vasopressin ([figure 1F](#)). Leukapheresis was performed on day 18 with moderate improvement in white cell count. Despite maximum cardiopulmonary support, the patient's condition continued to deteriorate, and he passed away on the same day. An autopsy confirmed hyperleukocytosis-driven, multi-organ, extensive vascular occlusion and infiltration by T and NK cells (antibody to distinguish NKT from T cells was not available in the pathology panel) ([figure 1G](#), top and middle). Evaluation of residual tumors showed ganglionic differentiation and absence of poorly differentiated neuroblasts, consistent with therapeutic response ([figure 1G](#), bottom).

During the initial CRS event, several cytokines were elevated compared with baseline (day -4) including IL-1 α / β , IL-2, IL-6, IL-7, IL-17 α , IL-9, IL-10, IFN γ , IL-15, MIP1 α / β , and TNF α . By day 14, the cytokine profile had changed with several previously elevated cytokines such as IL1 α / β , IL-2, IL-7, and IL-9 showing decreased concentrations while others including IL-6, IL-15, IFN γ , MIP1 α / β , and TNF α continued to increase. The highest absolute cytokine concentrations were detected for CXCL10, MCP1, TNF α , GRO, and MDC, and the greatest

fold-change for FLT-3L, IL-15, IL-6, MIP1 α , and TNF α ([figure 1H](#)). Several analytes including IL-15, CXCL10, TNF α , MIP α / β , IFN γ , and GM-CSF were detected at higher levels in this patient compared with mean concentrations in the first 12 patients ([figure 1I](#)).

Flow cytometry assessment of blood samples showed a rapid increase in CAR+ and CAR- populations of NKT, NK, and T cells beginning on day 14 ([figure 1J,K](#)). Notably, we observed a striking increase in CAR+ NK and γ / δ T cell numbers in day 14 peripheral blood mononuclear cells (PBMCs) compared with low levels in the pre-infusion product (online supplemental figure). Our analysis showed that this patient's infusion product had similar levels of NKT purity and CAR expression as well as populations of contaminating CAR+ T and NK cells compared with products from the 12 other patients (online supplemental figure 2A). However, the striking expansion of NKT, NK, and T cells that occurred in the peripheral blood of patient P5-1 was not observed in the other patients (online supplemental figure 2B,C). These results indicate robust polyclonal lymphocyte expansion post-infusion with the greatest proportional increase in CAR+ NK and γ / δ T cell populations.

Next-generation exome sequencing analysis of the pre-infusion product, peripheral blood, and skin samples did not identify any variants of clinical significance in 79 genes known to be associated with various cancers (online supplemental table 1), suggesting that the hyperleukocytosis that occurred in this patient was unlikely the result of a known underlying genetic disorder. To assess clonality based on gamma-retroviral transgene integration sites, pre-infusion cells and PBMCs collected on days 14 and 17 were assessed using the SonicLength method at the University of Pennsylvania Viral/Molecular High Density Sequencing Core.^{4,6} The pre-infusion product showed a high degree of heterogeneity with >1000 unique transgene integration sites. As the lymphocyte population that expanded the most, the CAR+ NK subset was determined to be non-clonal as the clone with the most frequent transgene integration sites was detected at only <0.6% (online supplemental figure 3A). Additionally, CAR+ NKT and CAR+ T subsets did not show evidence of clonal expansion (online supplemental figure 3B,C), thus excluding the possibility that the hyperleukocytosis event was driven by expansion of clonal cells. To determine whether IL-15 secretion by cells in the pre-infusion product contributed to this event, we challenged the products with tumor cells in vitro and found that P5-1 cells produced comparable levels of IL-15 (average 84.27 pg/mL/ 10^5 cells, SD 3) to healthy donor NKTs transduced with the same construct as previously reported (range 83.3–177.6 pg/mL/ 10^5 cells),⁷ suggesting that IL-15 overproduction was unlikely the cause of the aggressive cell expansion in patient P5-1.

Patient P5-1 was the first to be treated on DL5 and the first to receive CAR NKTs that underwent secondary stimulation with K562-based A4-4 artificial antigen-presenting cells (aAPCs) instead of the NKT-depleted fraction of autologous PBMCs; all other culture conditions remained

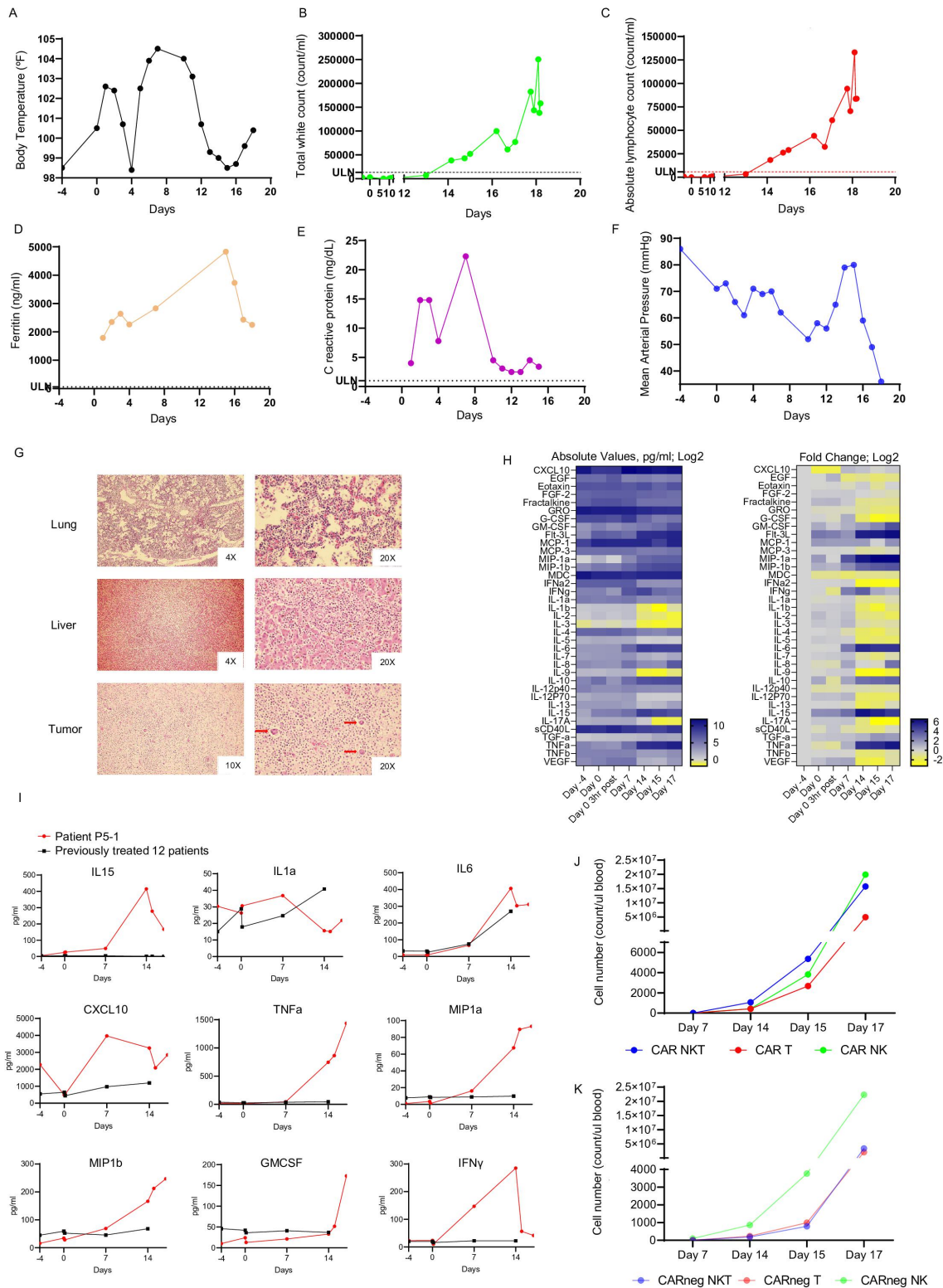


Figure 1 Patient P5-1 clinical findings and laboratory correlates. Clinical measurements were recorded for patient P5-1 from the start of lymphodepletion (day -4) through death (day 18). (A) Body temperature (°F). (B) Total white cell count (count/mL). (C) Absolute lymphocyte count (count/mL). (D) Ferritin level (ng/mL). (E) C reactive protein level (mg/dL). (F) Mean arterial pressure (mm Hg). (G) H&E staining of sections from patient P5-1 lung, liver, and tumor tissues at 4x and 20x magnification. Red arrows indicate residual tumor cells that have undergone ganglionic differentiation. (H) Blood draws were performed on patient P5-1 at indicated timepoints and serum cytokine levels were evaluated by Luminex assay. Absolute values (log₂ pg/mL) and fold change to baseline (log₂) of detected cytokines are shown. (I) Kinetics of indicated cytokine concentrations detected in patient P5-1 serum (red) from the start of lymphodepletion (day -4) through death (day 18) versus the average of 12 previously treated GINAKIT2 patients (black). (J, K) Kinetics of CAR+ and CAR- populations of NKT, T, and NK cells detected in patient P5-1 peripheral blood by flow cytometry analysis at indicated timepoints. IL, interleukin; NKT, natural killer T cells.

the same between the two protocols. The manufacturing protocol (online supplemental file 4) was originally amended to use aAPCs to ensure the presence of sufficient stimulatory cells for large numbers of NKTs and to provide optimal stimulation for superior NKT functionality and antitumor activity as demonstrated in preclinical models.⁸ To evaluate the potential for autonomous growth, cells from the pre-infusion product of patient P5-1 were thawed and cultured for 3 weeks without additional antigen stimulation or exogenous cytokines. The cells underwent a burst of expansion and growth stabilized after day 14 (figure 2A). We also observed an outgrowth of NK cells after 11 days in culture (online supplemental figure 4).

Such expansion without exogenous cytokine support was not observed in pre-infusion products from previously treated patients that underwent either primary or primary and secondary stimulations with autologous PBMCs. However, three other products that were manufactured with aAPC secondary stimulation but have not been infused did undergo expansion in this assay (figure 2A). In one case, two different products were manufactured from a single patient, one with aAPC and one with PBMC secondary stimulation; a burst of expansion was only observed in the aAPC-stimulated product (figure 2B).

Next, CAR NKT products were manufactured from 12 healthy donors using either PBMCs or aAPCs for secondary stimulation. Only aAPC-stimulated CAR NKTs significantly expanded ex vivo, and this expansion was observed with the cells that were cryopreserved at early but not later timepoints (2 or 4 days vs 10 days post-stimulation) (figure 2C, online supplemental figure 5). Additionally, IL-15 levels were significantly higher in the culture of aAPC-stimulated versus PBMC-stimulated CAR NKTs (figure 2D). Collectively, these results suggest that using aAPCs for secondary stimulation of CAR NKTs contributes to transient ex vivo proliferation in the absence of exogenous cytokines, which may have enabled the extensive hyperleukocytosis seen in this patient.

Single-cell RNA sequencing of the aAPC-stimulated pre-infusion product from patient P5-1 revealed distinct gene expression clusters compared with products that underwent secondary stimulation with autologous PBMCs (patients P1-1, P1-2, P2-3, and P3-1). P5-1 cells were depleted for expression of genes in clusters 1 (NKT2 in G1 phase) and 2 (NKT2 mitotic phase) and enriched for genes in clusters 3 (NKT1 mitotic phase), 6 (NKT17 mitotic phase), and 7 (NKT1 S phase)⁹ (figure 2E, online supplemental table 2). This expression pattern differed from that of patients P1-1, P1-2, P2-3, and P3-1, whose products were more similar to each other. Cell cycle scoring indicated a bimodal distribution of cells in S phase for patient P5-1, with subpopulations of cells in clusters 3, 6, and 7 expressing high levels of proliferation-associated genes (figure 2F). Higher expression levels of BATF3 and TYMS in patient P5-1's product indicate a memory-like phenotype and greater proliferative potential (online

supplemental figure 6A). Additionally, lower levels of CD69 and IL2RA and higher levels of HLA-DRA and HLA-DRB5 suggest a later stage of activation in patient P5-1's cells versus other patients (online supplemental figure 6B,C). These findings show that patient P5-1's product has a distinct gene expression profile consistent with high proliferative potential and sustained activation.

DISCUSSION

We report a case of a NB patient who developed hyperleukocytosis after GD2-CAR.15 NKT administration on a phase 1 clinical trial. Our root-cause analysis revealed no genetic alterations of known clinical significance and excluded the possibility of clonal expansion due to insertional mutagenesis. Based on our findings that aAPC- but not PBMC-stimulated CAR NKTs derived from both NB patients and healthy donors undergo a burst of ex vivo expansion when cultured in the absence of exogenous cytokines, we propose that stimulation of NKT and NK cells with A4-4 aAPCs may have led to stochastic activation of lymphocytes in patient P5-1's product such that an initial burst of CAR NK/NKT proliferation was generated. This proliferation was potentially amplified following in vivo exposure to tumor antigen, resulting in a positive feedback loop of expansion and production of transgenic IL-15, which on reaching a critical systemic level fueled further expansion of IL-15-sensitive NK, γ/δ T, CD8 T, and NKT cells regardless of CAR expression.

The A4-4 clone was derived from K562 and specifically selected for stimulation of CAR NKT products manufactured for this trial based on its ability to mediate optimal stimulation and expansion of human NKT cells through expression of CD1d^{med}, CD86^{high}, 4-1BBL^{med}, and OX40L^{high}.⁸ K562-based aAPCs have been employed in other studies to expand multiple types of immune effector cells including NK and T cells,^{10,11} and this is the first clinical observation of potential aAPC-associated lymphoproliferation and lethal toxicity. The singularity of this event among the many patients who have been infused with aAPC-activated T or NK cell products suggests that it may be unique to this specific patient. Additionally, cytokine armoring is a common strategy for enhancing the efficacy of T- and NK-based therapeutics against multiple malignancies,^{12,13} and it warrants careful preclinical and clinical monitoring for severe toxicities. For example, CAR NK cells expressing IL-15 have been reported to cause toxicity in preclinical AML models.¹⁴ However, IL-15 levels secreted by patient P5-1's product measured within an acceptable range and therefore cannot explain the triggering event that led to initial expansion of the infused cells. Instead, after accumulating systemically, IL-15 likely contributed at a later stage of this event by supporting the growth of both CAR+ and CAR- lymphocytes in patient P5-1, leading to uncontrolled lymphocytosis and toxicity. Alternatively, this hyperleukocytosis event may have been associated with a metapneumovirus infection that occurred in the patient at the time of CAR NKT infusion

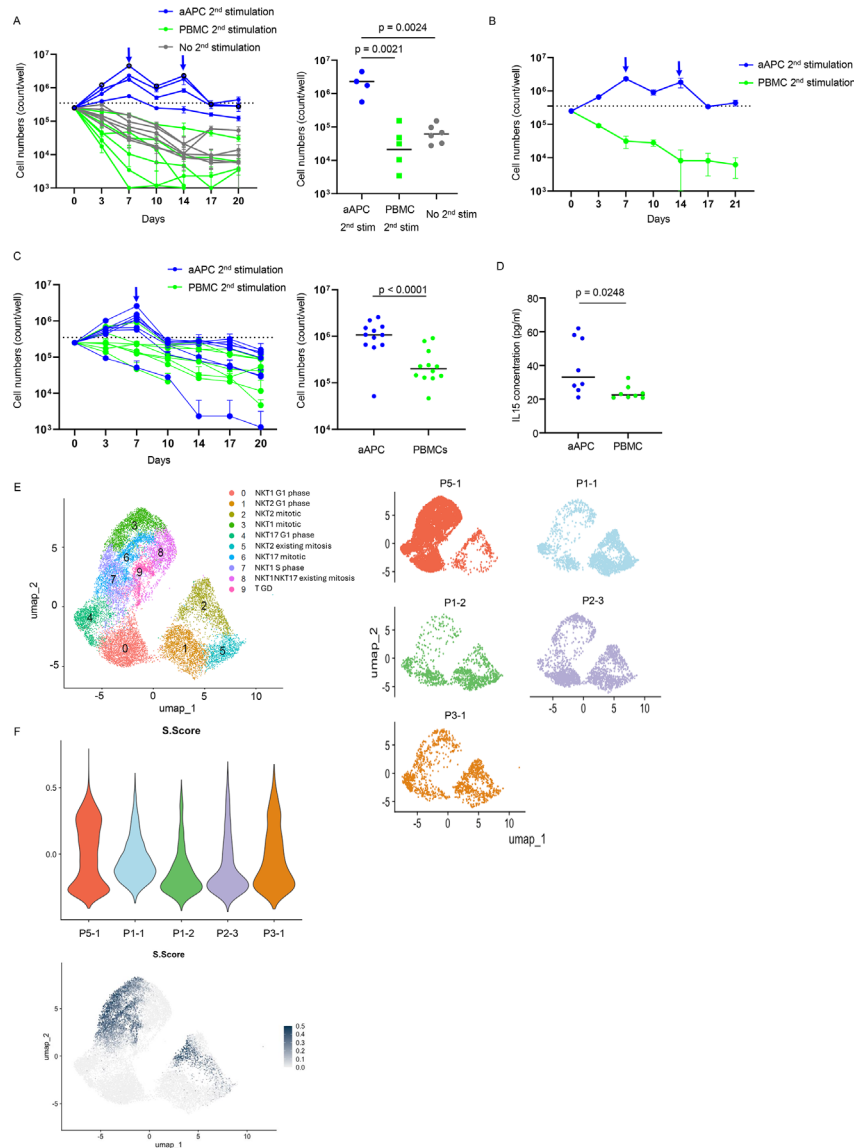


Figure 2 Secondary stimulation of GD2-CAR.15 NKTs with A4-4 aAPCs versus PBMCs leads to significant differences in autonomous growth and gene expression patterns. (A) Thawed GD2-CAR.15 NKT cells from all GINAKIT2 patients were cultured in complete RPMI without exogenous cytokines for 21 days and cell numbers were calculated at indicated timepoints. aAPC secondary-stimulated products (blue lines, blue line with black dot shows patient P5-1) were cryopreserved at day 4–5 post-secondary stimulation (day 5 for P5-1). PBMC secondary-stimulated products (green lines) were cryopreserved on days 3–5 post-secondary stimulation. Products that did not undergo secondary stimulation were cryopreserved on days 9–11 of expansion (gray lines). Blue arrows at days 7 and 14 indicate timepoints when cells were diluted to 2.5×10^5 cells/well starting concentration when numbers exceeded 3.75×10^5 threshold (dotted line). Graph on right shows statistical analysis of day 7 cell counts with indicated p values, one-way analysis of variance, Tukey's multiple comparisons. (B) GD2-CAR.15 NKT cells from a different GINAKIT2 patient underwent secondary stimulation with either A4-4 aAPCs or PBMCs. Thawed products were cultured for 21 days as in (A). Growth kinetics for aAPC-stimulated (blue line) and PBMC-stimulated (green line) products during this assay are shown. (C) GD2-CAR.15 NKTs were generated from 12 healthy donors, secondary-stimulated with either A4-4 aAPCs or PBMCs, and cryopreserved on day 4 post-secondary stimulation. Thawed products were cultured for 21 days as in (A). Growth kinetics for aAPC-stimulated (blue lines) and PBMC-stimulated (green lines) products during this assay (left); statistical analysis of day 7 counts, $p < 0.0001$, Student's t-test (right). Blue arrow at day 7 indicates when cells were diluted to 2.5×10^5 cells/well starting concentration because numbers exceeded 3.75×10^5 threshold (dotted line). (D) On day 7 of ex vivo culture without exogenous cytokines shown in (C), IL-15 concentrations in the supernatants of products from 8 of the 12 healthy donors were measured by ELISA, $p = 0.0248$, paired Student's t-test. (E) Single-cell RNAseq data from patient P5-1 (secondary stimulation with aAPCs) and patients P1-1, P1-2, P2-3, and P3-1 (secondary stimulation with PBMCs) was analyzed to generate a UMAP visualization of differential subset distributions for 16,793 CAR+ cells. UMAPs of GD2-CAR.15 NKT product cells for individual patients are shown on the right. (F) Differential expression of proliferation-related genes in patient P5-1 and patients P1-1, P1-2, P2-3, and P3-1. Violin plot showing S-phase gene module score expression in indicated patient products (top). UMAP representation of GD2-CAR.15 NKT product cells colored by S-phase gene module score (bottom). aAPCs, artificial antigen-presenting cells; PBMCs, peripheral blood mononuclear cells.

or an undetected mutation that predisposed the patient to higher IL-15 sensitivity.

Importantly, we did detect evidence of therapeutic response in residual tumors collected from patient P5-1 at autopsy. Our findings warrant the development of measures to control cell activation during manufacture and to regulate or pharmacologically control transgene expression in CAR-based immunotherapy products, particularly those that are cytokine-armed, such that potential clinical benefit can be achieved without toxicity. We found that culturing CAR NKTs for several additional days post-stimulation diminishes the aAPC-induced proliferative boost. Similarly, introduction of “rest” through treatment with the multi-kinase inhibitor dasatinib during manufacture has been reported to control CAR T cell overactivation and exhaustion, increasing therapeutic potential without toxicity.¹⁵ After infusion, safety switches such as inducible caspase 9 can be used to control or eliminate therapeutic cells as needed.¹⁶ Moreover, the next generation of CAR/cytokine constructs may benefit from synthetic regulatory circuits designed to keep engineered immune cell numbers and function within an optimal range to maximize therapeutic potency and minimize toxicity.

To address the findings in this case report, we have developed a mitigation strategy that includes: (1) reinitiating autologous PBMCs for restimulation, (2) adding a modified ex vivo antigen-independent growth assay as a release criterion, and (3) increasing patient monitoring post-infusion. Following resumption of the clinical trial, we have treated four patients with PBMC-restimulated CAR NKTs without observing significant toxicity.

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Correction notice This article has been amended since it was first published online. To more concisely describe the course of events, a sentence in the third paragraph of the Case Description section has been changed to: ‘Despite maximum cardiopulmonary support, the patient’s condition continued to deteriorate, and he passed away on the same day’.

Acknowledgements The authors are grateful to M Brenner, H Heslop, and B Grilley who serve as IND sponsors on this study and provide regulatory support and personnel of the cGMP facility at the Center for Cell and Gene Therapy for manufacturing CAR-NKTs. We also thank P Srivaths, director of the Pheresis Service at Texas Children’s Hospital. We are thankful for the excellent technical assistance provided by the staff of the Flow Cytometry Core Laboratory at the Texas Children’s Cancer and Hematology Center and the Single Cell Genomics Core at Baylor College of Medicine.

Contributors AH and LSM designed the clinical trial. AH wrote the clinical protocol, and AH and GT are principal investigators of the clinical trial. AH, GT, and SB performed clinical evaluations. ANC, CMA, MSW and NG processed clinical samples

and performed flow cytometry and Luminex analyses. DS performed pathology analyses. DWP performed whole exome sequencing analyses. AM performed scRNA-seq and analyzed data. SR, AMR, and FDB performed insertion site analysis. GT, ANC, XX, GB, HY, and DdIC performed root cause analysis experiments and healthy donor studies. The manuscript was written by GT, ANC, AH, and LSM and edited by EJD. The guarantor of the study is LSM. All authors discussed and interpreted results.

Funding This work was supported by grants or contracts from Alex’s Lemonade Stand Foundation for Childhood Cancer, St. Baldrick’s Foundation, the American Cancer Society (to LSM and AH).

Competing interests No, there are no competing interests.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by the Baylor College of Medicine Institutional Review Board (BCM: H-41003). Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement All requests for raw and analyzed data and materials should be directed to LSM and will be promptly reviewed by the Baylor College of Medicine Licensing Group to verify if the request is subject to any intellectual property or confidentiality obligations. Patient-related data was generated as part of the clinical trial and may be subject to patient confidentiality. Any data and materials that can be shared will be released via a material transfer agreement. All raw data from single-cell sequencing are deposited in the Gene Expression Omnibus under accession number GSE274643 and previously uploaded dataset with accession number GSE223071.

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